

PROTEIN STRUCTURE, FUNCTION, AND SYNTHESIS

MODULE

5

The properties of amino acids will be discussed in this module followed by an examination of the four levels of protein structure. The process of translation where mRNA is used to synthesize proteins will be examined including a comparison of prokaryotes and eukaryotes. This will be followed by an overview of protein sorting in the cell.

Learning Outcomes:

- Describe the properties of amino acids and how linear polymers of amino acids form three-dimensional structures of proteins with specific functions.
- Describe the process of translation to create a newly synthesized peptide in prokaryotes and eukaryotes.
- Explain the Central Dogma of molecular biology, the role of mRNA and tRNA to the process of translation.
- Explain how protein synthesis relates to protein sorting in the cell.

✧ PROPERTIES OF AMINO ACIDS

A. Introduction

- Proteins are ubiquitous, diverse, and versatile
- Many of the cellular activities and reactions that are involved in cellular function are mediated by proteins
- Recall that proteins are linear polymers of a combination of 20 amino acids → like letters in an alphabet

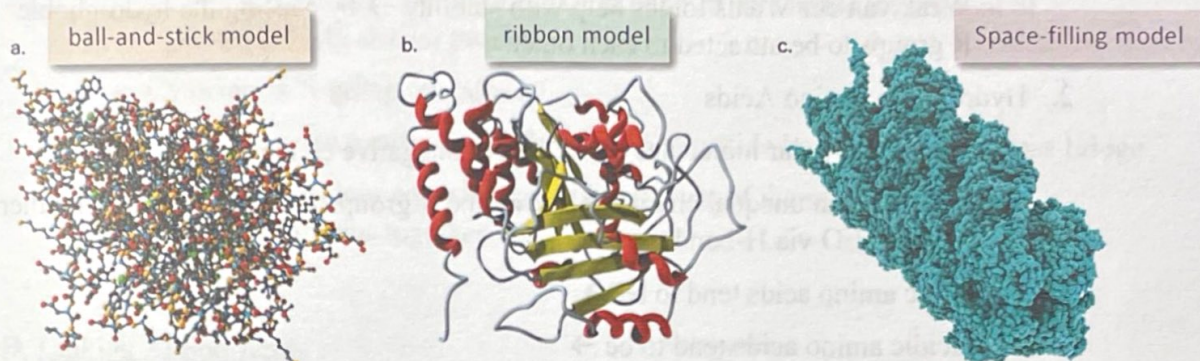


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Figure 5.1 Three Models of Protein Structure (Tubulin).

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B. Amino Acids

- Recall that amino acids are the building blocks of proteins → this is key as protein function is related to structure
- There are four components of an amino acid
 - Each component is bound to the alpha (α) carbon
 - Three components are identical for all amino acids:
 - a. the **carboxyl** group
 - b. the **amino** group
 - c. a **hydrogen**
 - At physiological pH, the amino & carboxyl groups are charged
- The fourth component is the side chain →
 - The R groups make each amino acid unique
 - Are responsible for the chemical and physical properties of each amino acid monomer

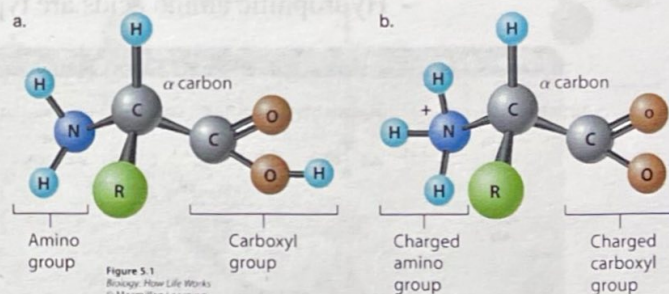


Figure 5.2 Amino Acid Structure.

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C. Amino Acid Classification

- The R groups of amino acids are grouped according to their properties
 - Polarity → polar or nonpolar
 - Interaction with water → hydrophilic or hydrophobic
 - Charge at physiological pH → acidic (-ve), basic (+ve), or neutral (uncharged)

1. Hydrophobic Amino Acids

- Hydrophobic amino acids tend to be buried in the interior of folded proteins
- The hydrophobic R groups aggregate together away from the water
- Weak van der Waals forces help with stability → by causing the hydrophobic R groups to be attracted to each other

2. Hydrophilic Amino Acids

- Recall that polar molecules contain electronegative elements →
- Results in an unequal charge → allows the R groups to interact with each other or with H₂O via H-bonding
- Basic amino acids tend to be →
- Acidic amino acids tend to be →
- Charged groups can form ionic bonds → with one another & with other charged molecules
- Hydrophilic amino acids are typically found on the “outer” surface of proteins



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Figure 5.3 Amino Acids.

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3. Special Amino Acids

- Three amino acids stand out because of how they affect the structure of proteins
- a. Glycine →
 - The α carbon is bonded to two H atoms → not an asymmetric carbon
 - It is small and nonpolar → allows free rotation around the C–N bond
 - This increases the flexibility of the polypeptide backbone
- b. Proline → R group links back to the amino group
 - Linkage restricts rotation of the C–N bond → this limits the amount of protein folding around proline
- c. Cysteine → R group contains a
 - This allows two cysteines to form a S–S disulfide bond → forms a cross-bridge
 - The cross-bridges can connect different parts of the same protein or different proteins together

D. Linking Amino Acids

- Adjacent amino acids are joined together in a peptide bond
- This is a dehydration reaction where the carboxyl group of one amino acid reacts with the amino group of another amino acid
 - Releases a molecule of H_2O
- The free amino group is at the amino end of the peptide → forms the
- The carboxyl group is at the carboxyl end → forms the
- A polymer of amino acids connected by peptide bonds is a polypeptide → used synonymously with protein

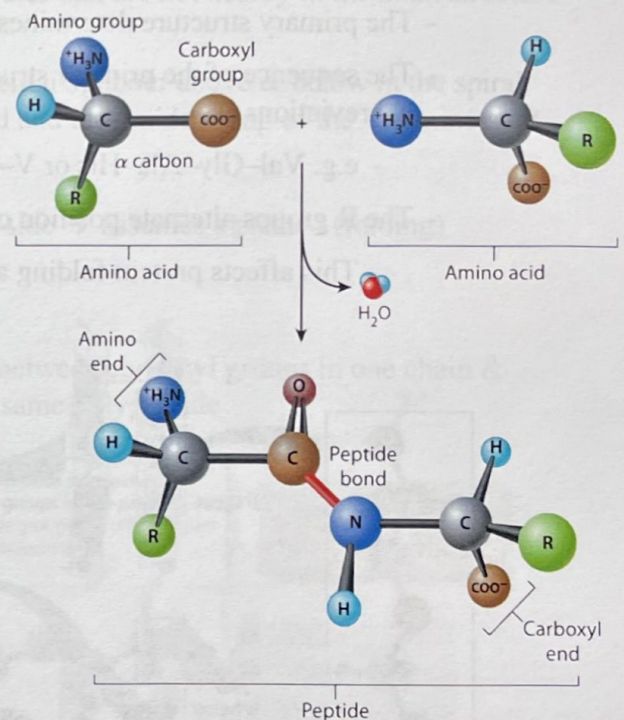


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Figure 5.4 Peptide Bond Formation.

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PROTEIN STRUCTURE

A. Introduction

- Each amino acid in a protein gives the structure and thus the function of the protein
 - The proteins will fold in a particular way based on the sequence of amino acids and thus the order of the R groups
- Protein structure has four levels of organization → primary (1°), secondary (2°), tertiary (3°), and quaternary (2°)
- The three-dimensional structure of a protein is the protein conformation → this is described by 2°, 3°, and 4°

B. Primary Structure

- The primary structure is the specific linear sequence of amino acids that make up the polypeptide chain → from amino end to carboxyl end
- Most polypeptides contain many amino acids → these are coded for by the genome
- The primary structure determines 2°, 3°, and 4° structure of the protein
- The sequence of the primary structure can be written as a sequence of the three letter or one-letter abbreviations
 - e.g. Val-Gly-Ala-His or V-G-A-H
- The R groups alternate position on either side of the chain of amino acids
 - This affects protein folding and the interaction of R groups

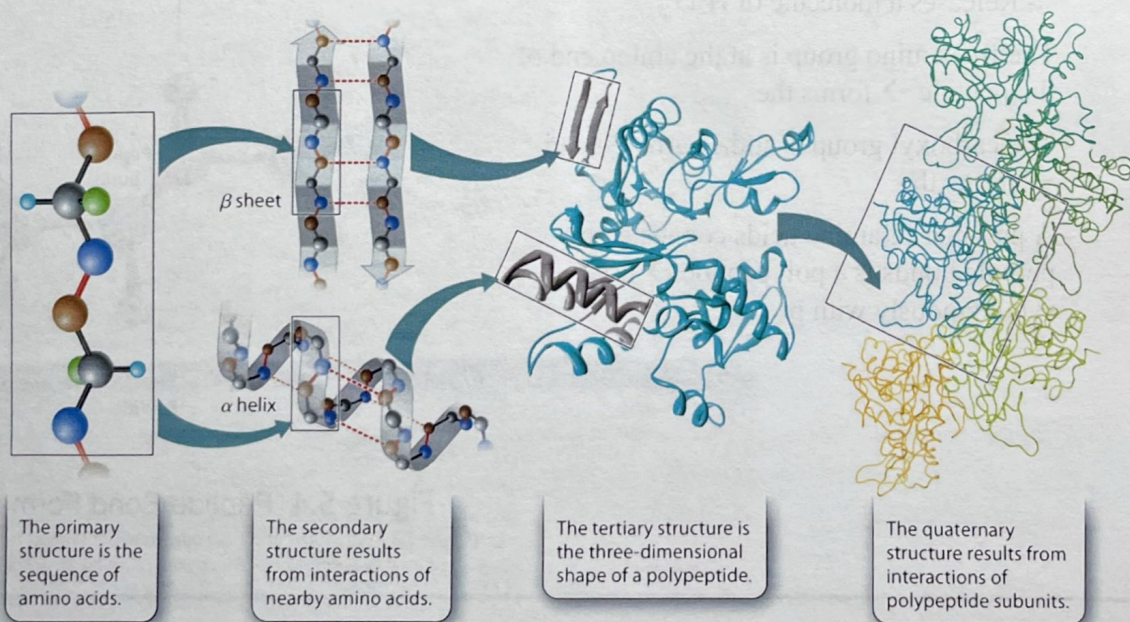


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Figure 5.5 Levels of Protein Structure.

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C. Secondary Structure

- Secondary structure describes the conformation of portions of the polypeptide chain
- Two types →
- Results from hydrogen bonding between neighbouring amino acids of the polypeptide backbone → occur between functional groups

- R groups are not involved

- It is a fixed configuration of the polypeptide backbone

i. Alpha Helix

- The alpha (α) helix is a very stable structure
- They are right-handed helices → let molecules that are not nearby in the main structure to interact with one another
- Form due to H-bonds to the 4th amino acid neighbour above & below in the spiral → the carbonyl group of one amino acid and the amide group of the 4th amino acid

ii. Beta Sheet

- Segments of the polypeptide lying side-by-side → assumes a pleated (folding) conformation
- Can be parallel or antiparallel
- Structure is stabilized by H-bonds formed between carbonyl groups in one chain & amide groups in the other chain within the same polypeptide

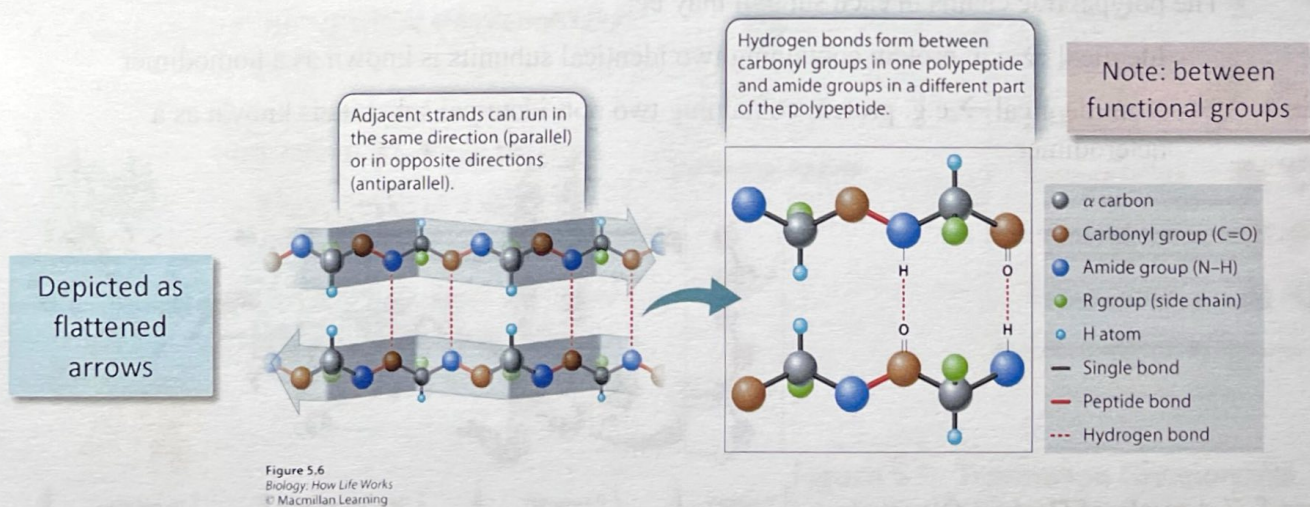


Figure 5.6 Alpha Helix (top) and Beta Sheet (bottom) Structure.

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D. Tertiary Structure

- The tertiary structure describes the conformation of the entire protein → this is a single polypeptide chain folded into 3° structure
- Ultimately this is how regions of 2° conformations are oriented
- Results in its functional form → unless it is part of a protein that has multiple subunits
- Tertiary structure is determined by the following:
 - Spatial distribution of the hydrophilic and hydrophobic R groups
 - Chemical bonds and interactions that form between the R groups
 - Includes:
 - Hydrogen bonds, hydrophobic bonds & ionic bonds
 - Disulfide bonds → covalent bond between two cysteine residues
- Recall that the structure of a protein determines its function
 - The sequence of the primary structure governs the secondary and tertiary structure of a protein
 - The overall shape (3° structure) of a functional protein may result in
 - Areas of the protein that form active sites for enzymes
 - Exterior R groups that may impact how a protein interacts with other molecules or proteins

E. Quaternary Structure

- Many proteins are made up of more than one polypeptide chain →
- The spatial arrangement of these subunits is the 4° structure
- These arise due to the same bonds as found in 3° structure
- The polypeptide chains in each subunit may be:
 - Identical → e.g. protein containing two identical subunits is known as a homodimer
 - Non-identical → e.g. protein containing two non-identical subunits is known as a heterodimer

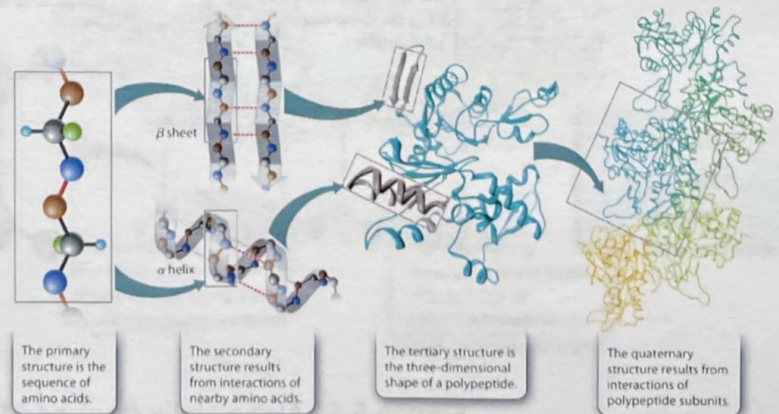


Figure 5.7 Levels of Protein Structure.

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OVERVIEW

A. Introduction

- Recall, the Central Dogma
- Transcription - sequence of DNA is used as a template
 - Makes the mRNA
- Translation - sequence of bases in mRNA is used to specify the order of amino acids to be added in the growing polypeptide
 - Final stage of the Central Dogma

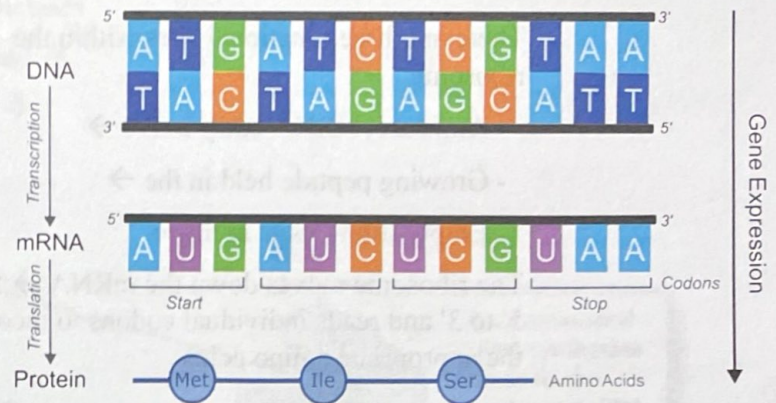


Figure 5.8 The Central Dogma.

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B. Translation

- Translation requires several components:
 - mRNA
 - Ribosome
 - Transfer RNAs →
 - Aminoacyl tRNA synthetases
 - Initiation factors, elongation factors & release factors

C. Ribosomes

- Ribosomes →
 - This is where translation takes place
- Ribosomes are a complex structure of RNA and protein
 - They consist of a small subunit and a large subunit
- Eukaryotic ribosomes are larger than prokaryotic ribosomes

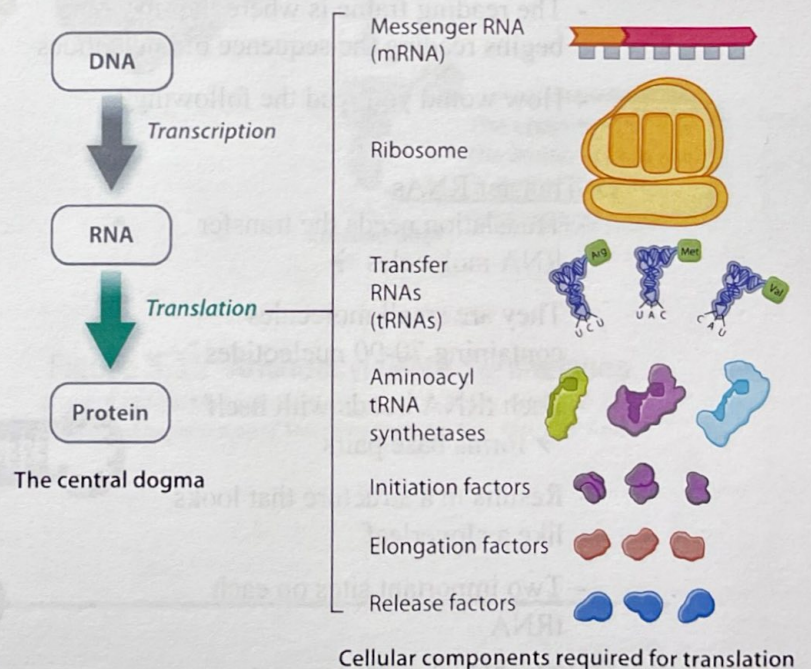


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Figure 5.9 Translation Components.

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- The mRNA is bound by the large and small ribosomal subunits → it then moves through the center of the ribosome
- There are three functional sites within the ribosome
 - Aminoacyl tRNA entry in the →
 - Growing peptide held in the →
 - Empty tRNA exits in the →
- The ribosome moves down the mRNA → from 5' to 3' and reads individual codons to incorporate the appropriate amino acids

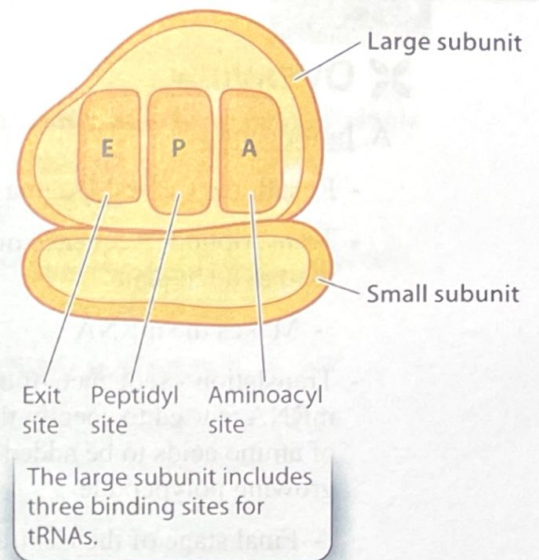


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Figure 5.10 Ribosome Structure.

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Question: What is a codon?

Answer: A codon is a nucleotide combination that specifies the placement of an amino acid → codes for the amino acid placement

- Each group of three adjacent nucleotides → arranged as a nonoverlapping series of nucleotide triplets
- The reading frame is where the ribosome begins reading the sequence of nucleotides
- How would you read the following?

ZWT HEB IGB OYS AWT HEM ANR UN

Z WTH EBI GBO YSA WTH EMA NRU N

ZW THE BIG BOY SAW THE MAN RUN

ZWTHEBIGBOYSAWTHEMANRUN

D. Transfer RNAs

- Translation needs the transfer RNA molecules →
- They are small molecules containing 70-90 nucleotides
- Each tRNA bonds with itself → forms base pairs
- Results in a structure that looks like a cloverleaf
- Two important sites on each tRNA
 - The 3' hydroxyl site on the 5'-CCA-3' end of the tRNA → where the specific amino acid attaches
- Three bases in the anticodon loop make up the →

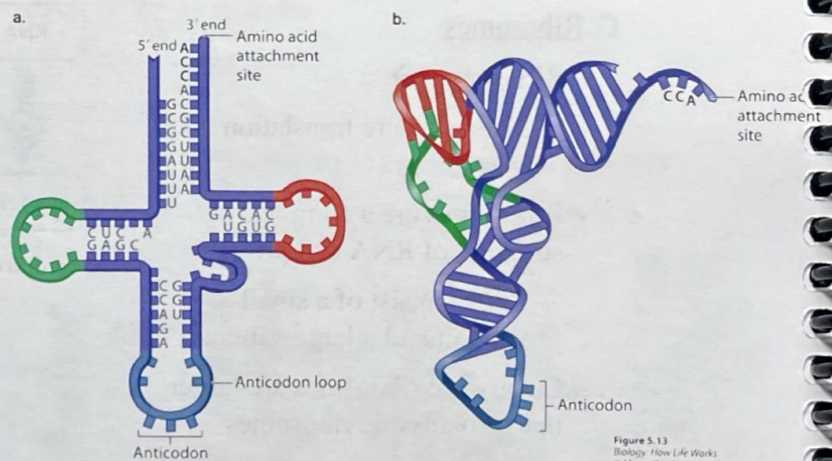


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Figure 5.11 Transfer RNA.

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Question: How are amino acids attached to the tRNA?

Answer: Specific amino acids are connected to specific tRNA molecules by enzymes called → aminoacyl tRNA synthetases

- A tRNA without an amino acid attached →
- A tRNA with an amino acid attached →
- tRNA synthetases are very accurate

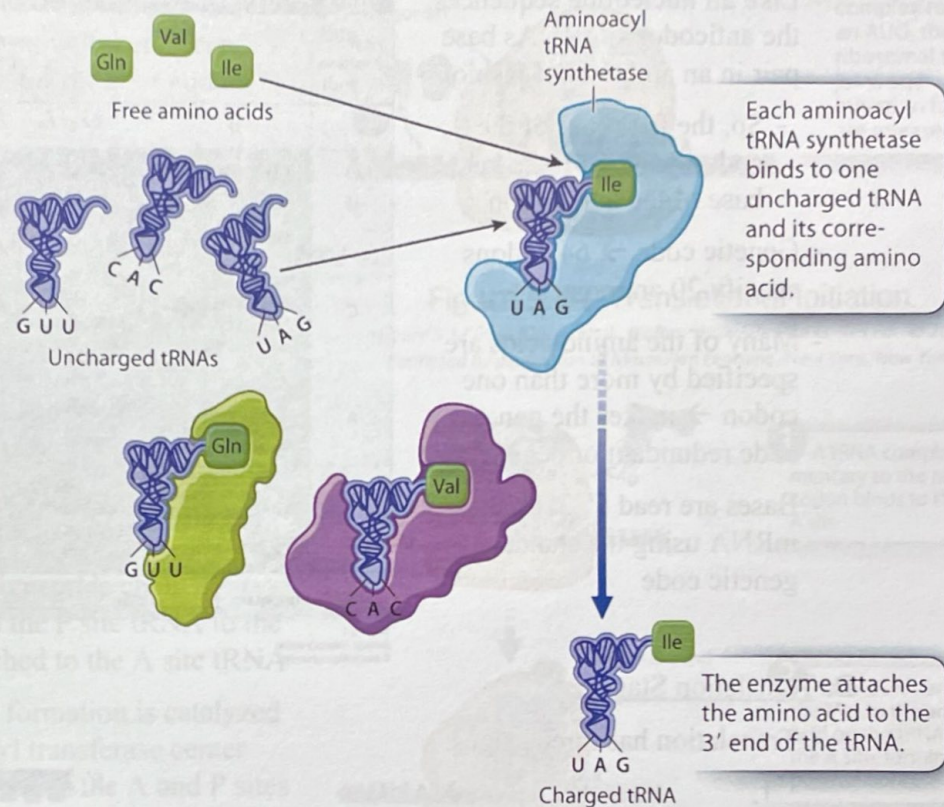


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Figure 5.12 Aminoacyl tRNA Synthetases.

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TRANSLATION PROCESS

A. The Genetic Code

- During translation, the anticodon of the tRNA base pairs with the codon on the mRNA
 - Recall base pairing is the specificity of DNA-RNA or codon-anticodon interactions
- Like all nucleotide sequences, the anticodon of tRNAs base pair in an antiparallel fashion
 - So, the first base of the codon pairs with the last base of the anti-codon
- Genetic code → 64 codons specify 20 amino acids
- Many of the amino acids are specified by more than one codon → makes the genetic code **redundant** or **degenerate**
- Bases are read 5'–3' on the mRNA using the standard genetic code

Table 5.1 The Genetic Code.

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TABLE 5.1 The Standard Genetic Code.

First position (5' end)	Second position				Third position (3' end)
	U	C	A	G	
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA Stop UAG Stop	UGU Cys UGC Cys UGA Stop UGG Trp	U C A G
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg	U C A G
A	AUU Ile AUC Ile AUA Ile AUG Met	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg	U C A G
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly	U C A G

Legend: Nonpolar (green), Polar (blue), Basic (yellow), Acidic (pink), Stop codon (purple)

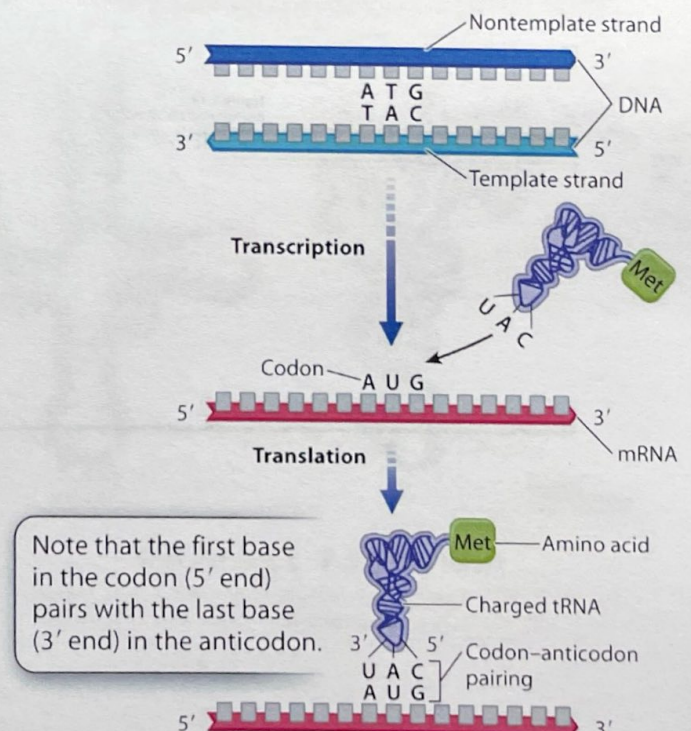
Table 5.1
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B. Translation Stages

- Translation has three stages:
 1. → AUG
codon is recognized & methionine is the first amino acid
 2. → each successive amino acid is added to the growing polypeptide chain
 3. → adding amino acids stops & the polypeptide chain is released from the ribosome

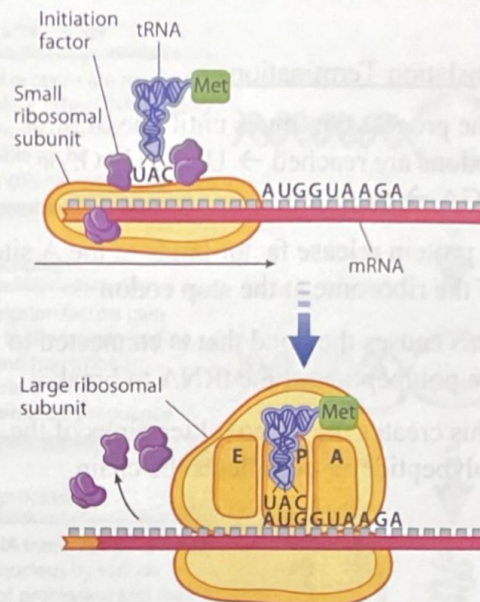
Figure 5.13 Role of Base Pairing.

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C. Eukaryotic Translation Initiation

- Stages 2 & 3 of translation are similar between prokaryotes & eukaryotes → however there are differences in step 1
- In eukaryotes the initiation complex forms at the 5' cap of the mRNA
 - The small ribosomal subunit and initiation factors will then “scan” the mRNA → for the start codon on the mRNA - AUG
- Once the start codon is reached the large ribosomal subunit is then recruited → translation can start
- The initiation factors are released



1 Initiation factors recruit the small ribosomal subunit and tRNA^{Met} and scan the mRNA for an AUG codon.

2 When the complex reaches an AUG, the large ribosomal subunit joins and initiation factors are released.

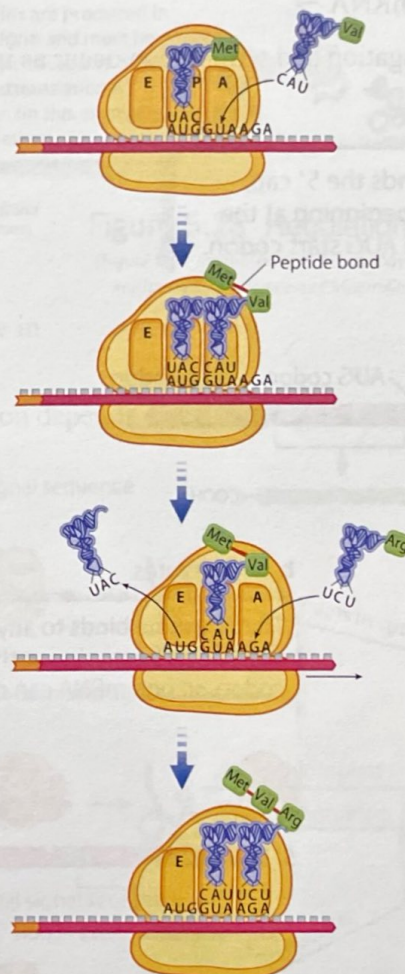
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Figure 5.14 Translational Initiation.

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D. Translation Elongation

- After ribosomal assembly, an aminoacyl-tRNA enters the **A site** and base-pairs with the next codon
- The growing polypeptide chain is transferred from the P site tRNA to the amino acid attached to the **A site** tRNA
 - Peptide bond formation is catalyzed by the peptidyl transferase center → located between the A and P sites of the large ribosomal subunit and composed of rRNA
- The ribosome then shifts one codon to the right
- This moves the uncharged tRNA (Met) to the **E site**
- The peptide-bearing tRNA moves to the P site
- This frees the **A site** for the next charged tRNA in line to enter based on the next codon



1 A tRNA complementary to the next codon binds to the A site

2 A reaction transfers the Met to the amino acid on the tRNA in the A site, forming a peptide bond.

3 The ribosome moves down one codon, which puts the tRNA carrying the polypeptide into the P site and the now-uncharged tRNA into the E site, where it is released. A new tRNA complementary to the next codon binds to the A site.

4 The polypeptide transfers to the amino acid on the tRNA in the A site. The polypeptide is elongated by repeating steps 3 and 4.

Figure 5.15 Translational Elongation.

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E. Translation Termination

- The process continues until one of three codons are reached → UAA, UAG, or UGA → these are
- A protein release factor binds to the A site of the ribosome at the stop codon
- This causes the bond that is connected to the polypeptide of the tRNA to break
- This creates the carboxyl terminus of the polypeptide → completes the chain

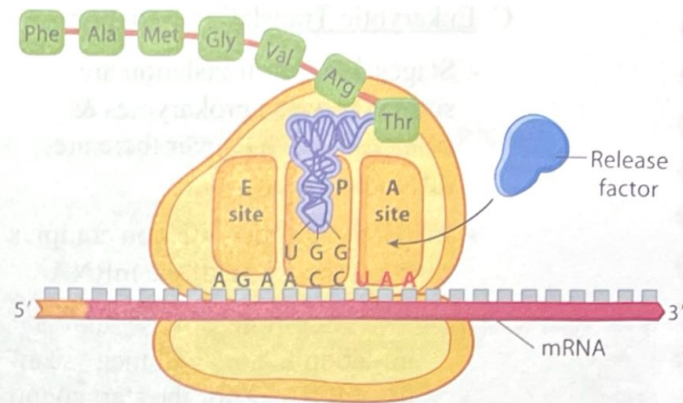


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Figure 5.16 Translational Termination.

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F. Prokaryotic Translation Initiation

- Recall the mRNA molecules have no 5' cap in prokaryotes
- Thus, the initiation complex is formed at one or more internal sequences present in the mRNA →
- Elongation and termination occur as in eukaryotes

a. Eukaryotes

The ribosome binds the 5' cap, with translation beginning at the first downstream AUG start codon.

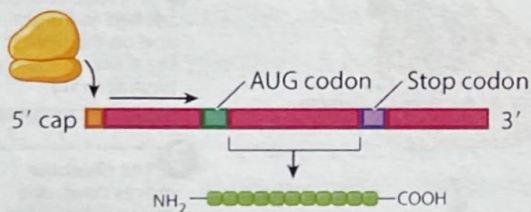


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b. Prokaryotes

The ribosome binds to any Shine–Dalgarno sequence, with translation beginning at the first downstream AUG start codon, so one mRNA can code for several polypeptides.

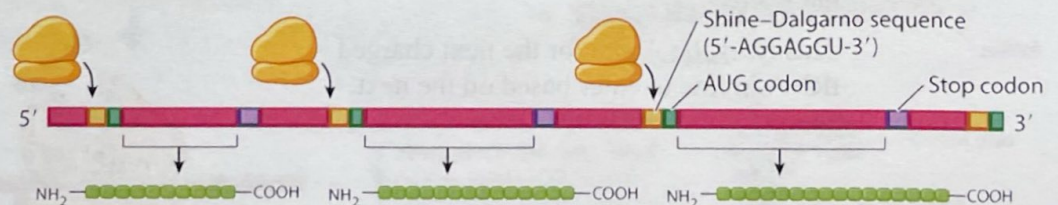


Figure 5.17 Translational In Eukaryotes & Prokaryotes.

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✱ POST TRANSLATION

A. Regulating Protein Synthesis

- Not all genes are expressed in the cell all the time
- There are many levels of regulation in the cell including:
 - DNA accessibility
 - Transcription factors
 - RNA processing
 - Post-translational modifications

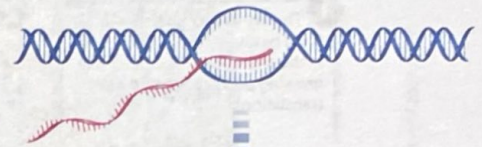
a. DNA accessibility

Special proteins are required to displace other DNA-binding proteins and make it accessible to the transcriptional machinery.



b. Transcription

Transcription factors gain access to the protein-coding gene and transcribe an RNA strand with the complementary sequence.



c. RNA processing

The RNA transcript is modified in the nucleus by various types of processing and then transported to the cytoplasm for translation into protein.



d. Posttranslational modification

Many proteins are produced in an inactive form and must be activated by cleavage or other types of posttranslational modification (in this example, phosphorylation).

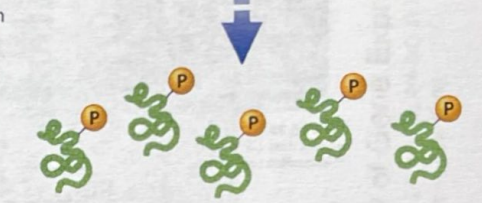


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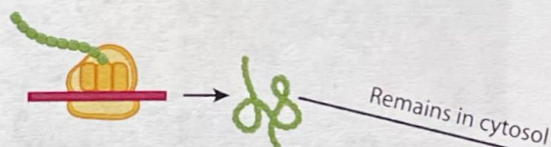
Figure 5.18 Regulation of Gene Expression.

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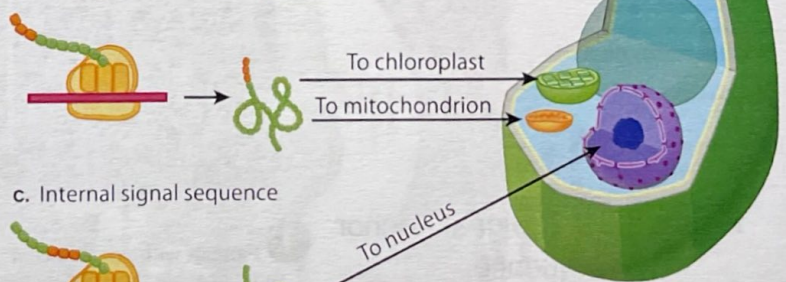
B. Protein Sorting

- Remember, mRNA is bound to a ribosome in the cytosol → where translation begins
- What happens to the protein after translation depends on how it will be sorted → specific signal sequences
 - No signal → stay in cytosol
 - Amino terminal signal → to chloroplast or mitochondria
 - Internal signal → to nucleus

a. No signal sequence



b. Amino-terminal signal sequence



c. Internal signal sequence



Figure 5.19 Signal Sequences on Proteins Synthesized by Free Ribosomes.

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- Some proteins have a specific amino-terminal signal sequence for transport to the ER
- This specific amino-terminal sequence signal is bound by the signal recognition particle →
- The ribosome with associated mRNA and newly formed polypeptide is transported to the ER

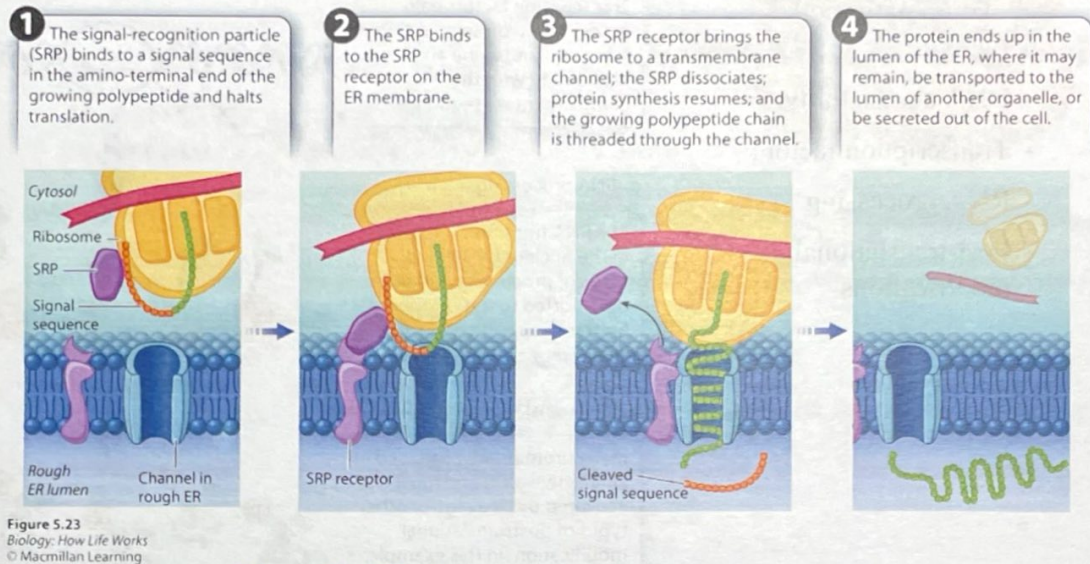


Figure 5.20 SRP & SRP Receptor Interaction.

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- Proteins produced by ribosomes on the rough ER could be found:
 - Embedded in the ER membrane → inserted as it is synthesized
 - Within the lumen of the endomembrane system
 - Secreted out of the cell

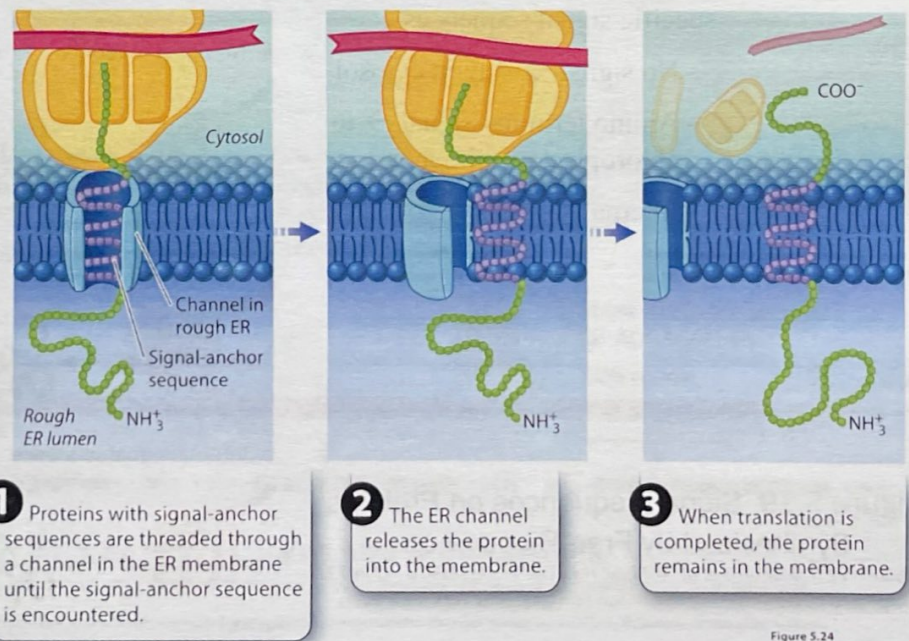


Figure 5.21 A Signal Anchor Sequence.

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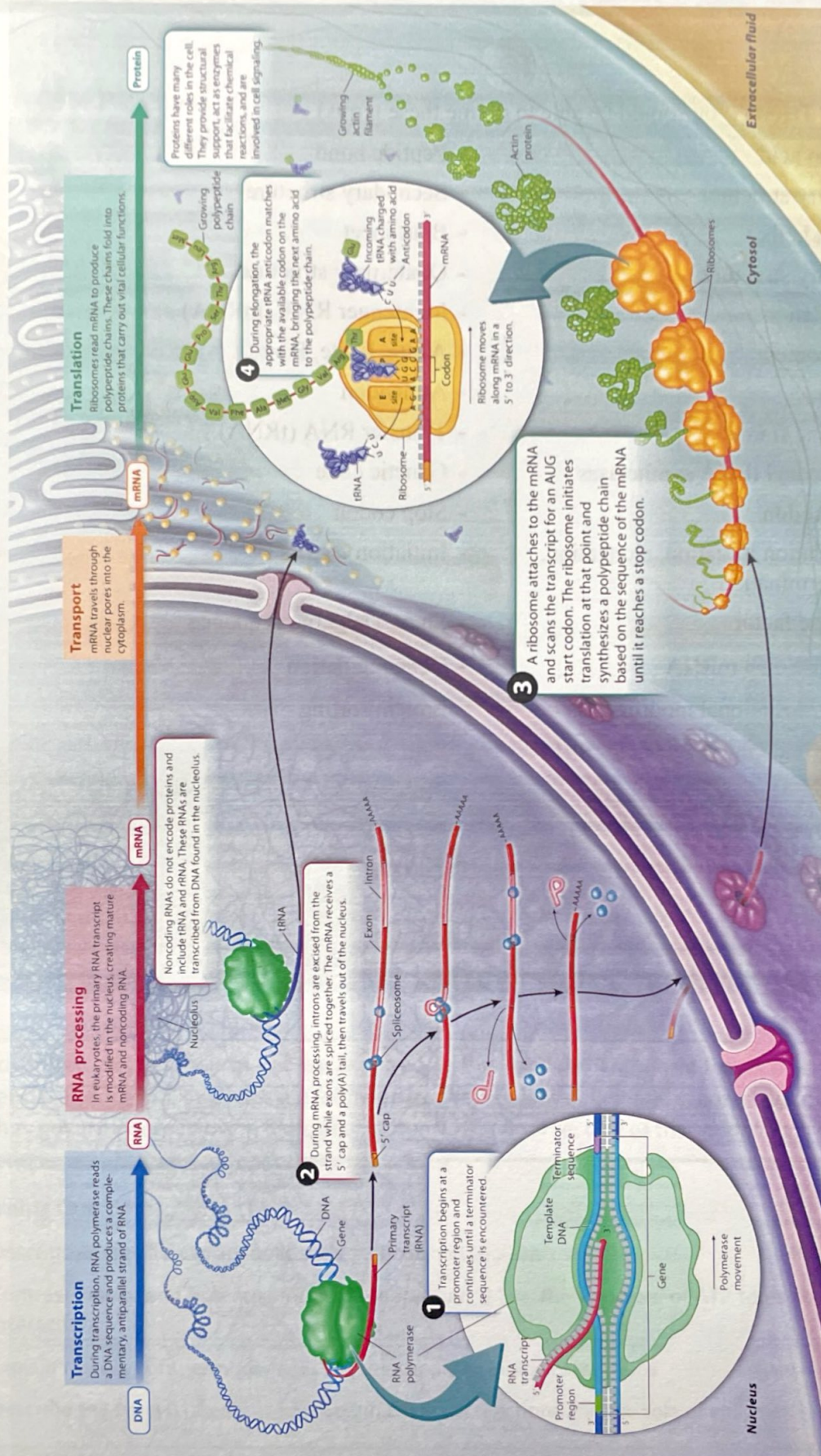


Figure 5.22 Overview of Gene Expression.

(Figure Visual Synthesis in Chapter 5 from Morris et al., *Biology: How Life Works*, 4th Ed., © 2023. Reprinted by permission of Macmillan Learning, New York, New York.)

IMPORTANT TERMS (you should be able to define these terms)

- Amino acid
- Primary structure
- Alpha helix
- Tertiary structure
- Translation
- Ribosomes
- Codon
- Reading frame
- Aminoacyl tRNA synthetases
- Start codon
- Translation initiation, elongation, and termination
- Release factor
- Polycistronic mRNA
- Post-translational modification
- Signal-recognition particle (SRP)
- Peptide bond
- Secondary structure
- Beta sheet
- Quaternary structure
- Messenger RNA (mRNA)
- A site, P site, E site
- Anticodon
- Transfer RNA (tRNA)
- Genetic code
- Stop codon
- Initiation factors
- Shine-Dalgarno sequence
- Gene regulation
- Protein sorting
- Signal sequences

IMPORTANT CONCEPTS (you should understand these concepts)

- Amino acid structure and properties of amino acids
- Molecular structure of proteins
- Protein synthesis
- Regulation of protein synthesis and sorting

MAKING LIFE WORK

MODULE

6

An overview of metabolism will be explored in this module. The basic forms of energy (kinetic and potential energy) will be discussed followed by a review of the first and second laws of thermodynamics as they govern energy flow in biological systems. The final section of the module will examine chemical reactions, emphasizing the function of enzymes.

Learning Outcomes:

- Describe energy flow in anabolic and catabolic metabolism
- Contrast kinetic and potential forms of energy and relate the structure of ATP to its role in cellular energy interactions
- Explain the laws of thermodynamics as they apply to biological systems
- Describe the type(s) of energy involved in chemical reactions and the role of enzymes.

INTRODUCTION

A. Why Do Cells Need Energy?

- Recall that cells need:
 - A way to separate the inside of the cell from the outside → the membrane
 - A way to encode/transmit information → nucleic acids & proteins
 - Energy to do the work of the cell which includes:
 - Movement, growth, pump ions, and perform reactions needed for cellular function, etc.

B. How Do Organisms Harvest Energy?

- Organisms can harvest energy from their environment in one of two ways:
 - Obtain energy from the sun →
 - Plants are the most common → obtain energy from the sun to convert CO_2 & water into sugar and O_2
 - Obtain energy from chemical compounds →
 - Animals are the most common chemotrophs → obtain energy by ingesting other organisms and breaking them down into CO_2 & water
- Organisms can further be classified in two ways based on their source of carbon:
 - If the organism can convert CO_2 into glucose since they make their own organic source of carbon →
 - If the organism is unable to convert CO_2 , they must ingest other organisms/molecules for a source of carbon →

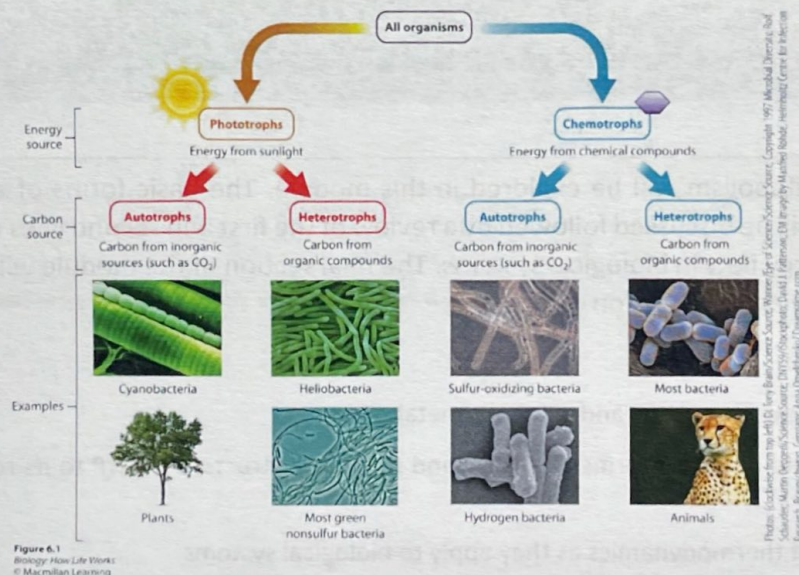


Figure 6.1 A Metabolic Classification of Organisms.

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METABOLISM

A. Introduction

Question: What is metabolism?

Answer: Metabolism → chemical processes that occur in a living organism to maintain life

- Includes the building up and breakdown of chemical compounds → allows energy to be harnessed or released
- These reactions are continuously happening in cells
- Many of these reactions are linked in such a way that the products of one reaction are the reactants of the next reaction →
- There are two branches of metabolism →

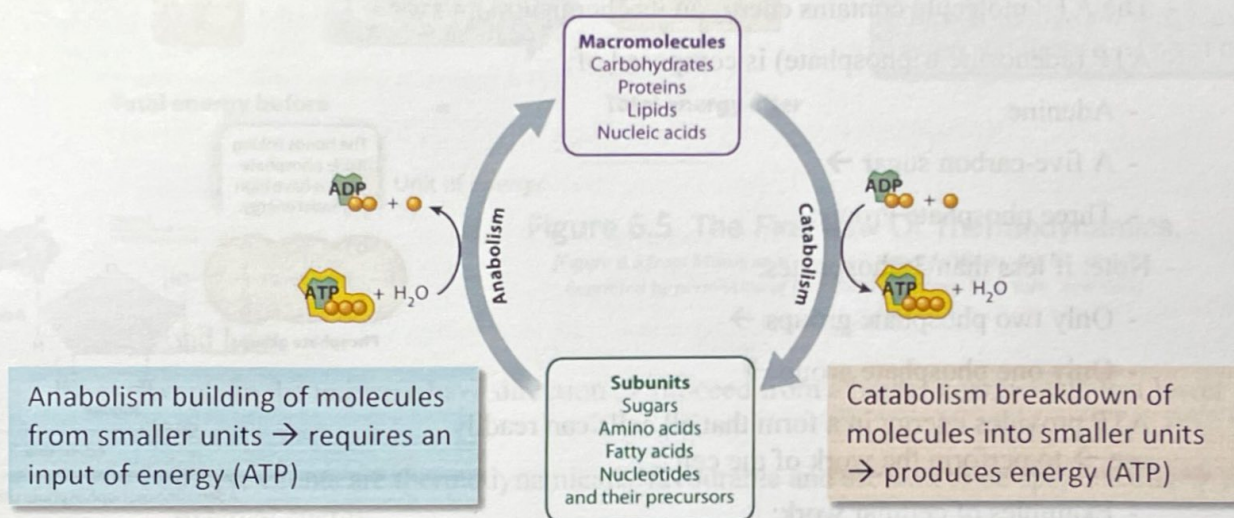


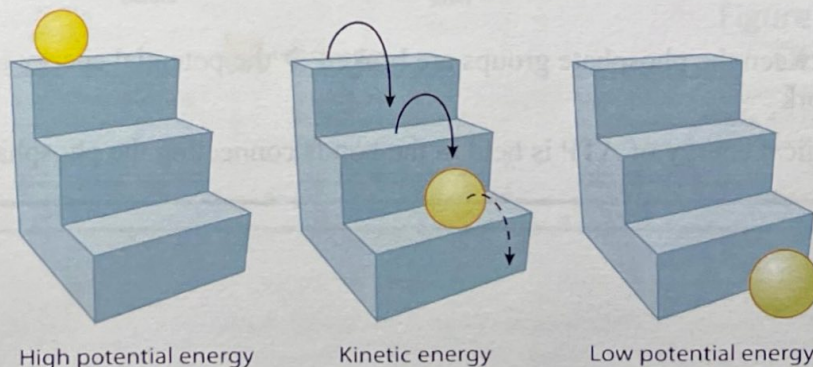
Figure 6.2
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Figure 6.2 Anabolism and Catabolism.

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B. Types of Energy

- Two types of energy:
 - Energy that is not associated with movement but rather is stored →
 - Energy of motion →



Note: energy can be converted from one form to another

Figure 6.3 Potential & Kinetic Energy.

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- Chemical bonds between pairs of atoms in a molecule hold onto a form of potential energy →
- Bond strength & potential energy are linked
 - Strong bonds = less potential energy
 - Weak bonds = more potential energy

C. ATP – The Energy Currency

- Cells do not use all the energy they have at once → best to keep it stored in a chemical form that is readily accessible to the cell
- Most common chemical form used by the cell is ATP →
- The ATP molecule contains energy in its chemical bonds
- **ATP** (adenosine triphosphate) is composed of:
 - Adenine
 - A five-carbon sugar →
 - Three phosphate groups
- Note: if less than 3 phosphates:
 - Only two phosphate groups →
 - Only one phosphate group →
- ATP provides energy in a form that all cells can readily use → to perform the work of the cell
 - Examples of cellular work:
 - DNA, RNA, and protein synthesis
 - Vesicle movement in a cell
 - Pumping substances across membranes
 - Breaking down fats to release energy
 - Breaking down proteins into amino acids
 - Oxidizing sugars to produce ATP
- ATP has a high amount of energy → the bonds between phosphate groups are weak
-
- When the bonds between the phosphate groups are broken → the potential energy is released for the cell to do work
- Important: the chemical energy of ATP is held in the bonds connecting the phosphate groups

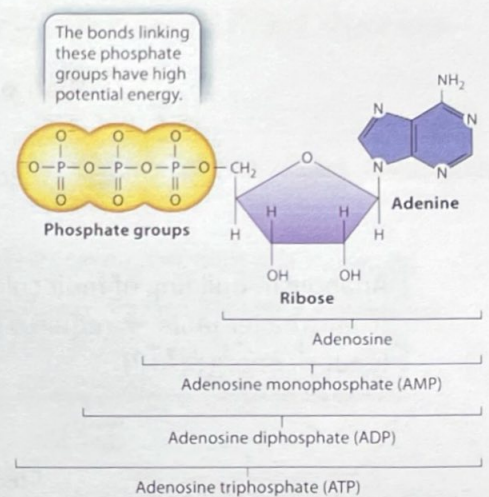


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Figure 6.4 ATP Structure.

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✧ LAWS OF THERMODYNAMICS

A. First Law

- Thermodynamics is the study of changes in energy that accompany events in the universe
- Concepts allow us to predict the direction that events will take and whether input of energy is required
 - First Law → energy can be neither created nor destroyed
- Energy can be converted from one form to another → e.g.

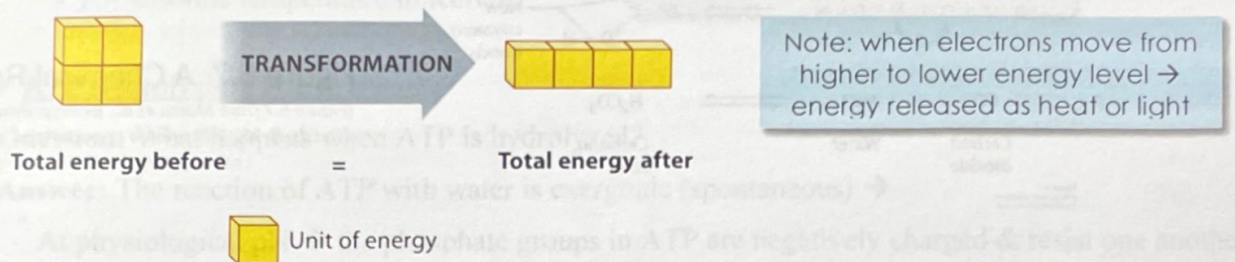


Figure 6.5 The First Law Of Thermodynamics.

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B. Second Law

- Events in the universe have direction → proceed from a higher energy state to a lower energy state
 - These events are thermodynamically favourable and are said to be spontaneous → no input of external energy
 - Results in a reduction in the amount of useable energy → energy transformations are not 100% efficient

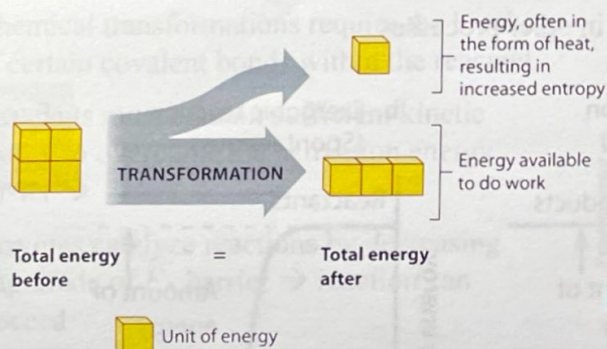


Figure 6.6 The Second Law Of Thermodynamics.

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CHEMICAL REACTIONS

A. Introduction

- Atoms keep their identity but the bonds linking them change during a chemical reaction
- The concentrations of products and reactants can influence whether a reaction proceeds forward or backward

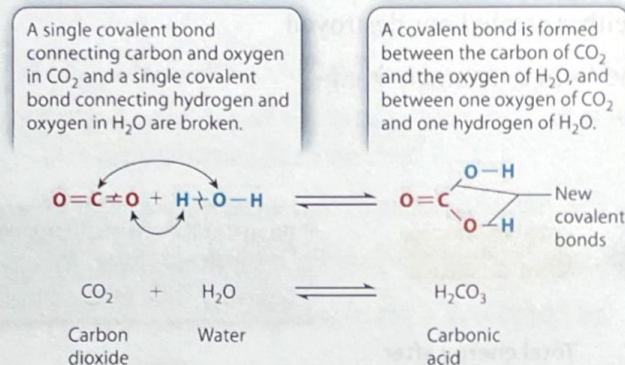


Figure 6.7 A Chemical Reaction.

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B. Free Energy (G)

- The amount of energy in a system available to do work → Gibbs free energy
 - ΔG is the symbol for the difference in Gibbs free energy between the reactants and products of a chemical process
- If the products of a reaction have more free energy than the reactants →
 - Thermodynamically unfavourable →
 - Needs an input of energy
- If the reactants have more free energy than the products →
 - Thermodynamically favourable →
 - Energy is available for use in other processes

Endergonic reactions
require energy

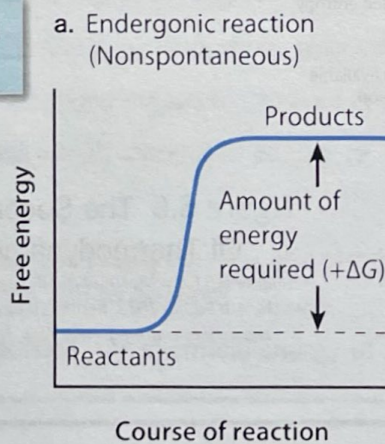
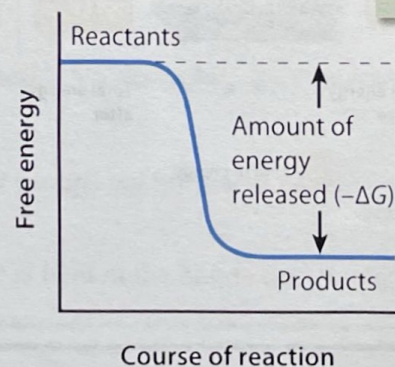


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b. Exergonic reaction
(Spontaneous)



Exergonic reactions
release energy

Figure 6.8 Endergonic and Exergonic Reaction.

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C. Energy Available

- Total energy (H) = energy available to do work (G) + energy lost to entropy (S)
- This can be rearranged to illustrate the state of Gibbs free energy:

$$\Delta G = \Delta H - T\Delta S$$

- Where:

- H = total energy (enthalpy)
- G = energy available to do work
- S = energy lost to entropy or disorder
- T = absolute temperature in Kelvin

Remember
 Δ = Change = Final State – Initial State

Temperature influences the movement of molecules & adds to the degree of disorder

D. ATP Hydrolysis

Question: What happens when ATP is hydrolyzed?

Answer: The reaction of ATP with water is exergonic (spontaneous) →

- At physiological pH → the phosphate groups in ATP are negatively charged & resist one another
- Compared to ATP, which has three phosphate groups, ADP contains two → meaning that it has a lower potential energy

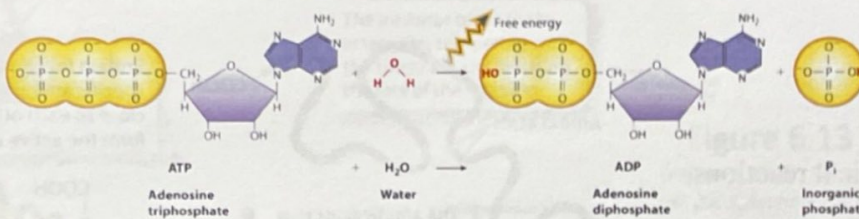


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Figure 6.9 ATP Hydrolysis.

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E. Activation Energy

- Chemical transformations require the breakage of certain covalent bonds within the reactant
- Reactants must contain sufficient kinetic energy to overcome the activation energy barrier →
- Enzymes catalyze reactions by decreasing magnitude of E_A barrier → reaction can proceed
- Cells can carry out many reactions with a $+\Delta G$

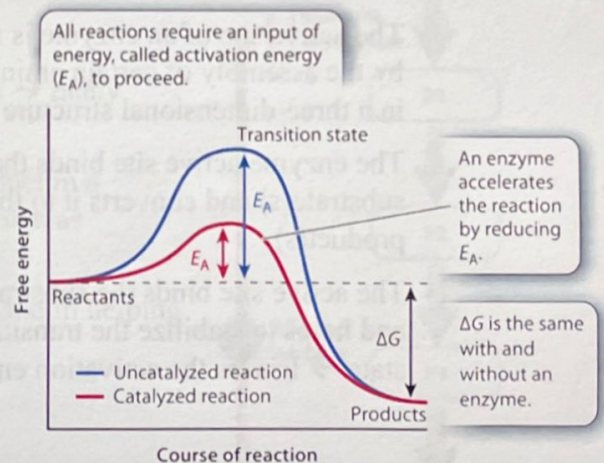


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Figure 6.10 Enzyme-catalyzed Reaction.

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- Occurs for two reasons:

1. Relative concentrations of reactants to products are maintained above that defined by equilibrium constant
2. Coupling endergonic and exergonic reactions
 - E.g. hydrolysis of ATP $\rightarrow$ drives endergonic reactions

The coupled reaction proceeds because the net ΔG is negative and P_i is shared between the two reactions.

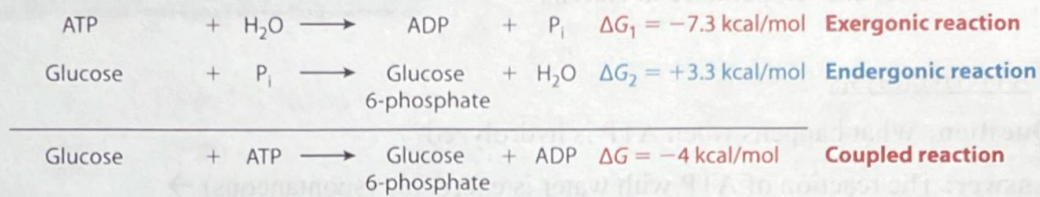


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Figure 6.11 Energetic coupling.

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F. Chemical Reactions

Question: How are chemical reactions catalyzed?

Answer: Chemical reactions are catalyzed by proteins called $\rightarrow$

- The active site of an enzyme is formed by the assembly of certain amino acids in a three-dimensional structure
- The enzyme active site binds the substrate(s) and converts it to the product(s)
- The active site binds the substrate(s) and helps to stabilize the transition state $\rightarrow$ lowers the activation energy

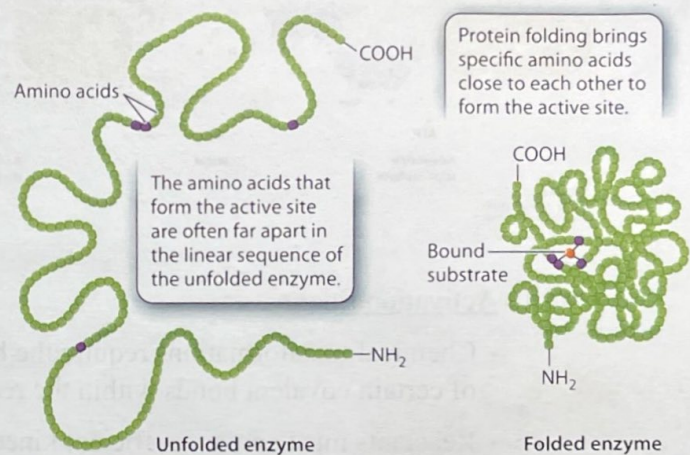


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a. Schematic model

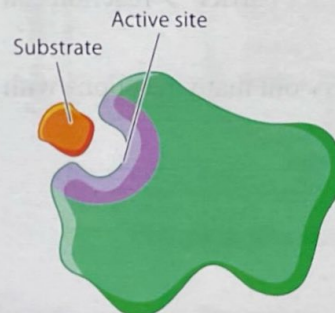


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b. Molecular model

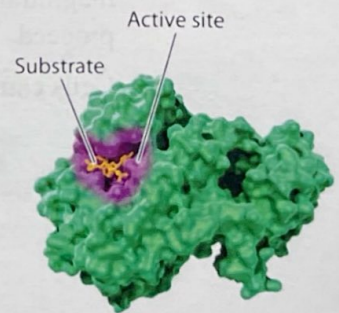


Figure 6.12 Enzyme Active Site.

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- Anabolic and catabolic reactions are catalyzed by enzymes
- It is possible to affect enzyme activity
 - Enzyme activity can be reduced by two different kinds of inhibitors → reversible & irreversible
 - Enzyme activity can be increased by activators

a. Lock-and-key model



b. Induced fit model

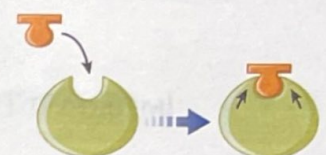
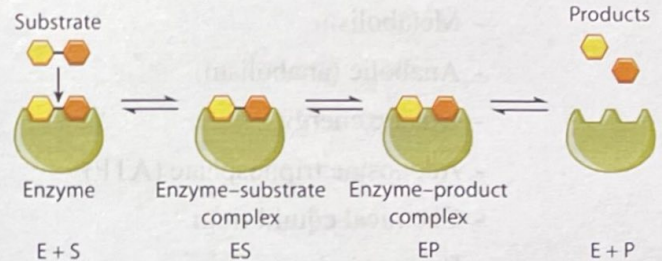


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a. Catabolic reaction



b. Anabolic reaction

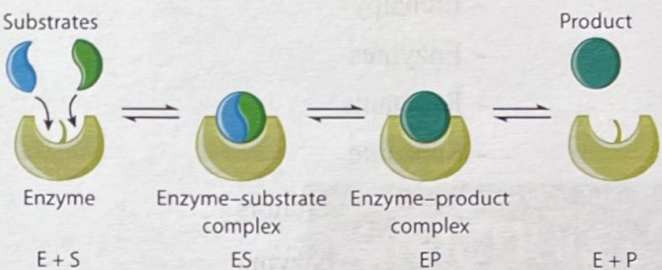
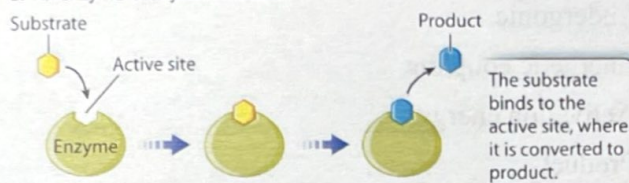


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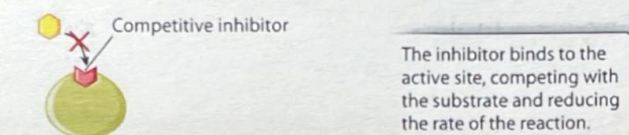
Figure 6.13 Enzyme-Substrate Interactions.

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a. An enzyme-catalyzed reaction



b. Competitive inhibition



c. Noncompetitive inhibition



- Because energy is needed for chemical reactions → many metabolic pathways are regulated
- An example of regulation is where the final product in a pathway inhibits the first step of the reaction known as →
- Important for maintaining balance in the cell and in helping the cell to conserve energy

Figure 6.14 An Allosteric Enzyme.

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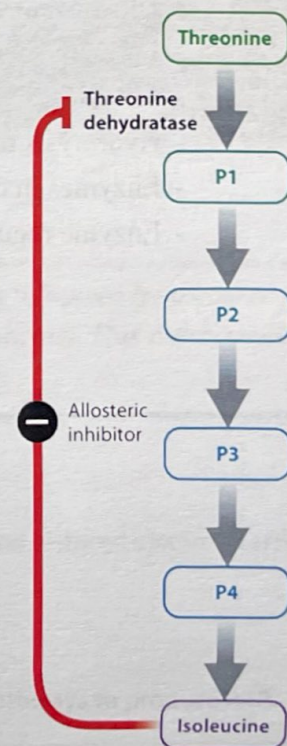


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IMPORTANT TERMS (you should be able to define these terms)

- Phototroph
- Autotroph
- Metabolism
- Anabolic (anabolism)
- Kinetic energy
- Adenosine triphosphate (ATP)
- Chemical equilibrium
- Exergonic
- Enthalpy
- Enzymes
- Reactants
- Substrate
- Enzyme inhibitors
- Allosteric enzyme
- Chemotroph
- Heterotroph
- Catabolic (catabolism)
- Potential energy
- Chemical energy
- Chemical reaction
- Gibbs Free Energy
- Endergonic
- Energetic coupling
- Activation energy
- Products
- Enzyme active site
- Enzyme activators

IMPORTANT CONCEPTS (you should understand these concepts)

- Overview of metabolism
- Kinetic and potential energy
- Laws of Thermodynamics
- Chemical reactions
- Hydrolysis of ATP
- Enzymes in catalyzing reactions
- Enzyme regulation